

KITABU QUEST

S3

Physics

Mascot: mountain gorilla

Scene: learners exploring physics with simple apparatus and a mountain gorilla mascot

CBC aligned draft learner book

Think • Explore • Achieve

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Title Page

KITABU QUEST S3 Physics

Rwanda REB-CBC learner book

Mascot: mountain gorilla

This KITABU QUEST book turns the curriculum into short lessons, worked examples, guided activities, practice, review, and home links for Rwanda learners.

How To Use This Book

This book helps S3 learners study Physics one teachable topic at a time.

1. Start with the inquiry question and say what you already know.
2. Read the Learn section slowly and mark new words.
3. Try the worked example before reading the full solution.
4. Complete the activity and practice tasks.
5. Correct your answer using the success criteria.
6. Ask for support when your answer is still unclear.

Subject method: Use measurements, diagrams, models, and step-by-step reasoning about systems and forces.

Learning Skills In This Book

Each topic builds knowledge, skill, values, communication, collaboration, critical thinking, creativity, digital awareness, and self-confidence.

Study actively: speak answers aloud, draw when useful, show your working, cite evidence, use local examples, and correct mistakes after feedback.

A strong answer is specific. It names the idea, gives evidence or method, applies it to the task, and checks whether it answers the question.

Table Of Contents

Use this table to move through the book one topic at a time. Each topic has a lesson opener, clear teaching, a model or example, guided work, practice, and checks for understanding.

1. Graphs of linear motion
2. Friction force and Newton's laws of motion
3. Applications of atmospheric pressure
4. Renewable and non-renewable energy sources
5. Heat transfer and quantity
6. Laws of thermodynamics
7. Electrical power transmission
8. Electric field intensity
9. House electric installation
10. Basic alternating current circuits
11. Refraction of light
12. Telecommunication Channels

As you finish each topic, return to the success criteria and correct one answer before moving on.

Graphs of linear motion: Lesson Opener

Learning focus: Graphs of linear motion

Country link: a safe Rwandan observation, model, data table, or investigation for Graphs of linear motion.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- plot and analyse the graphs of linear motion

Inquiry questions:

- How can graphs of linear motion help you solve a real problem in Physics?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Graphs of linear motion: Learn

Graphs of linear motion explains how position changes with time.

Core idea:

- Motion means an object changes position compared with a reference point.
- Distance tells how far an object moves.
- Time tells how long the motion takes.
- Speed compares distance and time.
- A distance-time table or graph can show whether motion is slow, fast, or changing.

Key words:

- Graphs of linear motion
- Graphs
- linear
- motion
- concept
- observation
- evidence
- safety

Graphs of linear motion: Worked Example

Worked example for Graphs of linear motion: A cart moves 12 metres in 4 seconds. Find its speed.

1. Speed = distance / time.
2. Distance = 12 m and time = 4 s.
3. $12 / 4 = 3$.

Answer: the cart moves at 3 m/s.

Graphs of linear motion: Vocabulary And Study Skill

Use these words and study moves before you practise Graphs of linear motion. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Graphs of linear motion
- Graphs
- linear
- motion
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Graphs of linear motion without a clear example, method, or evidence.

Graphs of linear motion: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Graphs of linear motion.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Graphs of linear motion: Practice And Correction

Practice for Graphs of linear motion:

Main outcome: plot and analyse the graphs of linear motion.

1. A bicycle moves 30 m in 6 s. Calculate its speed.
2. Draw a two-row table for distance and time.
3. Explain what happens to speed if the same distance takes more time.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Graphs of linear motion: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Graphs of linear motion.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Graphs of linear motion: Outcome Check

Outcome check for Graphs of linear motion: Plot and analyse the graphs of linear motion.

1. Calculate the speed of an object that moves 20 m in 5 s.
2. State the formula used.
3. Write the unit.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Friction force and Newton's laws of motion: Lesson Opener

Learning focus: Friction force and Newton's laws of motion

Country link: a safe Rwandan observation, model, data table, or investigation for Friction force and Newton's laws of motion.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- perform experiments involving Newton's laws of motion and friction force

Inquiry questions:

- How can friction force and newton's laws of motion help you solve a real problem in Physics?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Friction force and Newton's laws of motion: Learn

Friction force and Newton's laws of motion explains how forces affect motion using Newton's laws.

Core idea:

- First law: an object stays at rest or keeps moving in a straight line unless an unbalanced force acts on it.
- Second law: a larger force causes a larger acceleration, while a larger mass is harder to accelerate.
- Third law: when one object exerts a force on another, the second object exerts an equal and opposite force.
- Force diagrams help you show direction and balance.

Key words:

- Friction force and Newton's laws of motion
- Friction
- force
- Newton's
- laws
- motion
- concept
- observation

Friction force and Newton's laws of motion: Worked Example

Worked example for Friction force and Newton's laws of motion: Explain why a football at rest starts moving when kicked.

1. Before the kick, the ball is at rest.
2. The foot applies an unbalanced force on the ball.
3. Because of Newton's first law, the ball changes its state of motion only when the force acts.
4. The ball also pushes back on the foot with an equal and opposite force, showing Newton's third law.

Conclusion: the kick changes the motion because it provides an unbalanced force.

Friction force and Newton's laws of motion: Vocabulary And Study Skill

Use these words and study moves before you practise Friction force and Newton's laws of motion. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Friction force and Newton's laws of motion
- Friction
- force
- Newton's
- laws
- motion
- concept
- observation

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Friction force and Newton's laws of motion without a clear example, method, or evidence.

Friction force and Newton's laws of motion: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Friction force and Newton's laws of motion.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Friction force and Newton's laws of motion: Practice And Correction

Practice for Friction force and Newton's laws of motion:

Main outcome: perform experiments involving Newton's laws of motion and friction force.

1. State Newton's first, second, and third laws in your own words.
2. Draw a force diagram for a pushed box or kicked ball.
3. Explain which law is shown when the object starts moving, speeds up, or pushes back.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Friction force and Newton's laws of motion: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Friction force and Newton's laws of motion.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Friction force and Newton's laws of motion: Outcome Check

Outcome check for Friction force and Newton's laws of motion: Perform experiments involving Newton's laws of motion and friction force.

1. State one Newton law correctly.
2. Apply it to a moving or resting object.
3. Draw or describe the force direction.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Applications of atmospheric pressure: Lesson Opener

Learning focus: Applications of atmospheric pressure

Country link: a safe Rwandan observation, model, data table, or investigation for Applications of atmospheric pressure.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- explain the existence of pressure in gas and the application of atmospheric pressure

Inquiry questions:

- How can applications of atmospheric pressure help you solve a real problem in Physics?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Applications of atmospheric pressure: Learn

Applications of atmospheric pressure is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Applications of atmospheric pressure
- Applications
- atmospheric
- pressure
- concept
- observation
- evidence
- safety

Applications of atmospheric pressure: Worked Example

Investigation model for Applications of atmospheric pressure: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Applications of atmospheric pressure: Vocabulary And Study Skill

Use these words and study moves before you practise Applications of atmospheric pressure. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Applications of atmospheric pressure
- Applications
- atmospheric
- pressure
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Applications of atmospheric pressure without a clear example, method, or evidence.

Applications of atmospheric pressure: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Applications of atmospheric pressure.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Applications of atmospheric pressure: Practice And Correction

Practice for Applications of atmospheric pressure:

Main outcome: explain the existence of pressure in gas and the application of atmospheric pressure.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Applications of atmospheric pressure: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Applications of atmospheric pressure.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Applications of atmospheric pressure: Outcome Check

Outcome check for Applications of atmospheric pressure: Explain the existence of pressure in gas and the application of atmospheric pressure.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Renewable and non-renewable energy sources: Lesson Opener

Learning focus: Renewable and non-renewable energy sources

Country link: a safe Rwandan observation, model, data table, or investigation for Renewable and non-renewable energy sources.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- differentiate between renewable and non-renewable energy sources and give examples

Inquiry questions:

- How can renewable and non-renewable energy sources help you solve a real problem in Physics?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Renewable and non-renewable energy sources: Learn

Renewable and non-renewable energy sources connects work, power, and energy changes.

Core idea:

- Work is done when a force moves an object through a distance.
- Energy is the ability to do work.
- Power describes how quickly work is done or energy is transferred.
- Energy can change form, such as chemical to movement, electrical to light, or potential to kinetic.

Key words:

- Renewable and non-renewable energy sources
- Renewable
- non-renewable
- energy
- sources
- concept
- observation
- evidence

Renewable and non-renewable energy sources: Worked Example

Worked example for Renewable and non-renewable energy sources: A learner lifts a book from the floor to a desk.

1. The learner applies an upward force.
2. The book moves upward through a distance.
3. Work is done on the book.
4. The book gains gravitational potential energy.

Conclusion: work can transfer energy to an object.

Renewable and non-renewable energy sources: Vocabulary And Study Skill

Use these words and study moves before you practise Renewable and non-renewable energy sources. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Renewable and non-renewable energy sources
- Renewable
- non-renewable
- energy
- sources
- concept
- observation
- evidence

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Renewable and non-renewable energy sources without a clear example, method, or evidence.

Renewable and non-renewable energy sources: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Renewable and non-renewable energy sources.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Renewable and non-renewable energy sources: Practice And Correction

Practice for Renewable and non-renewable energy sources:

Main outcome: differentiate between renewable and non-renewable energy sources and give examples.

1. Give two examples of energy transformation.
2. Explain when work is done on an object.
3. Compare work and power in one sentence.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Renewable and non-renewable energy sources: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Renewable and non-renewable energy sources.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Renewable and non-renewable energy sources: Outcome Check

Outcome check for Renewable and non-renewable energy sources: Differentiate between renewable and non-renewable energy sources and give examples.

1. Give one example of energy changing form.
2. Explain when work is done.
3. Compare work and power in one sentence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Heat transfer and quantity: Lesson Opener

Learning focus: Heat transfer and quantity

Country link: a safe Rwandan observation, model, data table, or investigation for Heat transfer and quantity.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- evaluate modes of heat transfer and determine specific heat capacity of metal block

Inquiry questions:

- How can heat transfer and quantity help you solve a real problem in Physics?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Heat transfer and quantity: Learn

Heat transfer and quantity explains thermal energy and temperature.

Core idea:

- Temperature tells how hot or cold something is.
- Heat moves from a hotter object to a cooler object.
- Thermometers must be read carefully and safely.
- Different scales, such as Celsius and Kelvin, can describe temperature.

Key words:

- Heat transfer and quantity
- Heat
- transfer
- quantity
- concept
- observation
- evidence
- safety

Heat transfer and quantity: Worked Example

Investigation model for Heat transfer and quantity: Read water temperature safely.

1. Put warm water in a beaker under teacher guidance.
2. Place a thermometer bulb in the water without touching the bottom.
3. Wait for the reading to stop changing.
4. Read the Celsius scale at eye level.

Conclusion: careful thermometer use gives a reliable temperature.

Heat transfer and quantity: Vocabulary And Study Skill

Use these words and study moves before you practise Heat transfer and quantity. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Heat transfer and quantity
- Heat
- transfer
- quantity
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Heat transfer and quantity without a clear example, method, or evidence.

Heat transfer and quantity: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Heat transfer and quantity.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Heat transfer and quantity: Practice And Correction

Practice for Heat transfer and quantity:

Main outcome: evaluate modes of heat transfer and determine specific heat capacity of metal block.

1. Read a thermometer diagram or classroom thermometer safely.
2. State the direction heat flows between hot and cold objects.
3. Compare heat and temperature in one sentence.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Heat transfer and quantity: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Heat transfer and quantity.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Heat transfer and quantity: Outcome Check

Outcome check for Heat transfer and quantity: Evaluate modes of heat transfer and determine specific heat capacity of metal block.

1. State the direction heat flows.
2. Describe safe thermometer reading.
3. Use the word temperature correctly.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Laws of thermodynamics: Lesson Opener

Learning focus: Laws of thermodynamics

Country link: a safe Rwandan observation, model, data table, or investigation for Laws of thermodynamics.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- describe the internal energy of a system by applying laws of thermodynamics

Inquiry questions:

- How can laws of thermodynamics help you solve a real problem in Physics?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Laws of thermodynamics: Learn

Laws of thermodynamics is learned by asking a clear question, observing carefully, recording evidence, and explaining what the evidence means.

Core idea:

- Start with the object, organism, material, body part, force, energy change, or environmental feature being studied.
- Use safe observations, labelled diagrams, simple measurements, or comparison tables.
- Keep evidence separate from guesses: evidence is what you saw, measured, read from a table, or labelled correctly.
- A conclusion should answer the question and mention one limit or next observation.
- Local examples from home, school, farms, water sources, weather, health, and the environment make the science useful.

Key words:

- Laws of thermodynamics
- Laws
- thermodynamics
- concept
- observation
- evidence
- safety
- conclusion

Laws of thermodynamics: Worked Example

Investigation model for Laws of thermodynamics: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Laws of thermodynamics: Vocabulary And Study Skill

Use these words and study moves before you practise Laws of thermodynamics. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Laws of thermodynamics
- Laws
- thermodynamics
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Laws of thermodynamics without a clear example, method, or evidence.

Laws of thermodynamics: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Laws of thermodynamics.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Laws of thermodynamics: Practice And Correction

Practice for Laws of thermodynamics:

Main outcome: describe the internal energy of a system by applying laws of thermodynamics.

1. Write one clear science question for the topic.
2. Use a labelled diagram, observation table, or model to record evidence.
3. Write a conclusion that answers the question and uses topic vocabulary.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Laws of thermodynamics: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Laws of thermodynamics.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Laws of thermodynamics: Outcome Check

Outcome check for Laws of thermodynamics: Describe the internal energy of a system by applying laws of thermodynamics.

1. State the science question.
2. Show the evidence with a labelled diagram, table, model, or observation.
3. Write a conclusion that follows from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Electrical power transmission: Lesson Opener

Learning focus: Electrical power transmission

Country link: a safe Rwandan observation, model, data table, or investigation for Electrical power transmission.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- analyse the transmission of electrical power

Inquiry questions:

- How can electrical power transmission help you solve a real problem in Physics?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Electrical power transmission: Learn

Electrical power transmission connects work, power, and energy changes.

Core idea:

- Work is done when a force moves an object through a distance.
- Energy is the ability to do work.
- Power describes how quickly work is done or energy is transferred.
- Energy can change form, such as chemical to movement, electrical to light, or potential to kinetic.

Key words:

- Electrical power transmission
- Electrical
- power
- transmission
- concept
- observation
- evidence
- safety

Electrical power transmission: Worked Example

Worked example for Electrical power transmission: A learner lifts a book from the floor to a desk.

1. The learner applies an upward force.
2. The book moves upward through a distance.
3. Work is done on the book.
4. The book gains gravitational potential energy.

Conclusion: work can transfer energy to an object.

Electrical power transmission: Vocabulary And Study Skill

Use these words and study moves before you practise Electrical power transmission. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Electrical power transmission
- Electrical
- power
- transmission
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Electrical power transmission without a clear example, method, or evidence.

Electrical power transmission: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Electrical power transmission.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Electrical power transmission: Practice And Correction

Practice for Electrical power transmission:

Main outcome: analyse the transmission of electrical power.

1. Give two examples of energy transformation.
2. Explain when work is done on an object.
3. Compare work and power in one sentence.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Electrical power transmission: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Electrical power transmission.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Electrical power transmission: Outcome Check

Outcome check for Electrical power transmission: Analyse the transmission of electrical power.

1. Give one example of energy changing form.
2. Explain when work is done.
3. Compare work and power in one sentence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Electric field intensity: Lesson Opener

Learning focus: Electric field intensity

Country link: a safe Rwandan observation, model, data table, or investigation for Electric field intensity.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- calculate intensity of electric field due to one or more point charges

Inquiry questions:

- How can electric field intensity help you solve a real problem in Physics?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Electric field intensity: Learn

Electric field intensity is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Electric field intensity
- Electric
- field
- intensity
- concept
- observation
- evidence
- safety

Electric field intensity: Worked Example

Investigation model for Electric field intensity: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Electric field intensity: Vocabulary And Study Skill

Use these words and study moves before you practise Electric field intensity. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Electric field intensity
- Electric
- field
- intensity
- concept
- observation
- evidence
- safety

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Electric field intensity without a clear example, method, or evidence.

Electric field intensity: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Electric field intensity.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Electric field intensity: Practice And Correction

Practice for Electric field intensity:

Main outcome: calculate intensity of electric field due to one or more point charges.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Electric field intensity: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Electric field intensity.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Electric field intensity: Outcome Check

Outcome check for Electric field intensity: Calculate intensity of electric field due to one or more point charges.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

House electric installation: Lesson Opener

Learning focus: House electric installation

Country link: a safe Rwandan observation, model, data table, or investigation for House electric installation.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- analyse and carry out a simple electric installation

Inquiry questions:

- How can house electric installation help you solve a real problem in Physics?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

House electric installation: Learn

House electric installation is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- House electric installation
- House
- electric
- installation
- concept
- observation
- evidence
- safety

House electric installation: Worked Example

Investigation model for House electric installation: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

House electric installation: Vocabulary And Study Skill

Use these words and study moves before you practise House electric installation. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- House electric installation
- House
- electric
- installation
- concept

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about House electric installation without a clear example, method, or evidence.

House electric installation: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for House electric installation.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

House electric installation: Practice And Correction

Practice for House electric installation:

Main outcome: analyse and carry out a simple electric installation.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

House electric installation: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from House electric installation.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

House electric installation: Outcome Check

Outcome check for House electric installation: Analyse and carry out a simple electric installation.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Basic alternating current circuits: Lesson Opener

Learning focus: Basic alternating current circuits

Country link: a safe Rwandan observation, model, data table, or investigation for Basic alternating current circuits.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- design and analyse simple alternating current circuits

Inquiry questions:

- How can basic alternating current circuits help you solve a real problem in Physics?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Basic alternating current circuits: Learn

Basic alternating current circuits explains moving electric charge in circuits.

Core idea:

- A complete circuit is needed for current to flow.
- Cells, wires, switches, and bulbs can show current effects safely.
- Electricity can produce light, heat, sound, movement, and magnetism.
- Safety matters: never experiment with mains electricity.

Key words:

- Basic alternating current circuits
- Basic
- alternating
- current
- circuits
- concept
- observation
- evidence

Basic alternating current circuits: Worked Example

Circuit model for Basic alternating current circuits: Light a bulb safely.

1. Use a cell, two wires, a switch, and a small bulb.
2. Connect a complete circuit.
3. Close the switch and observe the bulb.
4. Open the switch and observe again.

Conclusion: current flows only when the circuit is complete.

Basic alternating current circuits: Vocabulary And Study Skill

Use these words and study moves before you practise Basic alternating current circuits. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Basic alternating current circuits
- Basic
- alternating
- current
- circuits
- concept
- observation
- evidence

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Basic alternating current circuits without a clear example, method, or evidence.

Basic alternating current circuits: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Basic alternating current circuits.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Basic alternating current circuits: Practice And Correction

Practice for Basic alternating current circuits:

Main outcome: design and analyse simple alternating current circuits.

1. Draw a complete circuit with a cell, switch, and bulb.
2. Explain why a broken circuit stops current.
3. List two electrical safety rules.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Basic alternating current circuits: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Basic alternating current circuits.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Basic alternating current circuits: Outcome Check

Outcome check for Basic alternating current circuits: Design and analyse simple alternating current circuits.

1. Draw a complete circuit.
2. Explain why a switch can stop current.
3. Write one electrical safety rule.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Refraction of light: Lesson Opener

Learning focus: Refraction of light

Country link: a safe Rwandan observation, model, data table, or investigation for Refraction of light.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- investigate and explain refraction of light using evidence

Inquiry questions:

- How can refraction of light help you solve a real problem in Physics?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Refraction of light: Learn

Refraction of light explains how light travels and reflects.

Core idea:

- Light travels in straight lines through a uniform medium.
- Shadows form when an opaque object blocks light.
- Reflection occurs when light bounces from a surface.
- Smooth, shiny surfaces reflect light more regularly than rough surfaces.

Key words:

- Refraction of light
- Refraction
- light
- concept
- observation
- evidence
- safety
- conclusion

Refraction of light: Worked Example

Investigation model for Refraction of light: Show that light travels in straight lines.

1. Make small holes in three cards.
2. Place a torch behind the first card.
3. Align the holes so light reaches a screen.
4. Move one card sideways and observe the screen.

Conclusion: the light path is blocked when the holes are not in a straight line.

Refraction of light: Vocabulary And Study Skill

Use these words and study moves before you practise Refraction of light. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Refraction of light
- Refraction
- light
- concept
- observation
- evidence
- safety
- conclusion

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Refraction of light without a clear example, method, or evidence.

Refraction of light: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Refraction of light.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Refraction of light: Practice And Correction

Practice for Refraction of light:

Main outcome: investigate and explain refraction of light using evidence.

1. Draw a straight ray of light from a torch.
2. Explain how a shadow forms.
3. Name one smooth reflector and one rough surface.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Refraction of light: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Refraction of light.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Refraction of light: Outcome Check

Outcome check for Refraction of light: Investigate and explain refraction of light using evidence.

1. Draw a straight light ray.
2. Explain how a shadow forms.
3. Describe reflection from a smooth surface.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Telecommunication Channels: Lesson Opener

Learning focus: Telecommunication Channels

Country link: a safe Rwandan observation, model, data table, or investigation for Telecommunication Channels.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- differentiate telecommunication channels

Inquiry questions:

- How can telecommunication channels help you solve a real problem in Physics?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Telecommunication Channels: Learn

Telecommunication Channels is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Telecommunication Channels
- Telecommunication
- Channels
- concept
- observation
- evidence
- safety
- conclusion

Telecommunication Channels: Worked Example

Investigation model for Telecommunication Channels: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Telecommunication Channels: Vocabulary And Study Skill

Use these words and study moves before you practise Telecommunication Channels. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Telecommunication Channels
- Telecommunication
- Channels
- concept
- observation

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Telecommunication Channels without a clear example, method, or evidence.

Telecommunication Channels: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Telecommunication Channels.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Telecommunication Channels: Practice And Correction

Practice for Telecommunication Channels:

Main outcome: differentiate telecommunication channels.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Telecommunication Channels: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Telecommunication Channels.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Telecommunication Channels: Outcome Check

Outcome check for Telecommunication Channels: Differentiate telecommunication channels.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Properties of Physical processes affecting plant growth: Lesson Opener

Learning focus: Properties of Physical processes affecting plant growth

Country link: a safe Rwandan observation, model, data table, or investigation for Properties of Physical processes affecting plant growth.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- describe the physical properties affecting plant growth

Inquiry questions:

- How can properties of physical processes affecting plant growth help you solve a real problem in Physics?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Properties of Physical processes affecting plant growth: Learn

Properties of Physical processes affecting plant growth teaches useful computer work through exact steps, not guessing.

Core idea:

- Word processing is used to create, edit, format, save, and print text documents.
- A spreadsheet organizes data in rows, columns, and cells.
- A cell can hold text, a number, or a formula.
- Formatting should make work clearer, not just more decorative.
- Good digital work is saved with a clear file name and checked before sharing.

Learner task link: create a class list, simple budget, marks table, timetable, or short report using safe school devices.

Key words:

- Properties of Physical processes affecting plant growth
- Properties
- Physical
- processes
- affecting
- plant
- growth
- concept

Properties of Physical processes affecting plant growth: Worked Example

Worked example for Properties of Physical processes affecting plant growth: Make a simple spreadsheet for class garden harvest.

1. Open a blank worksheet.
2. In row 1, type headings: Crop, Week 1, Week 2, Total.
3. Enter beans, maize, and tomatoes in the Crop column.
4. Type harvest numbers for Week 1 and Week 2.
5. In the Total column, use a formula such as =B2+C2.
6. Save the file with a clear name, for example garden-harvest-p6.

Quality check: headings are clear, numbers are in the correct cells, formulas give sensible totals, and the file is saved.

Properties of Physical processes affecting plant growth: Vocabulary And Study Skill

Use these words and study moves before you practise Properties of Physical processes affecting plant growth. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Properties of Physical processes affecting plant growth
- Properties
- Physical
- processes
- affecting
- plant
- growth
- concept

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Properties of Physical processes affecting plant growth without a clear example, method, or evidence.

Properties of Physical processes affecting plant growth: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Properties of Physical processes affecting plant growth.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Properties of Physical processes affecting plant growth: Practice And Correction

Practice for Properties of Physical processes affecting plant growth:

Main outcome: describe the physical properties affecting plant growth.

1. Create a document or worksheet with a clear title and headings.
2. Enter five rows of useful class, garden, timetable, budget, or marks data.
3. Format the work neatly, save it with a clear file name, and explain one formula or formatting choice.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Properties of Physical processes affecting plant growth: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Properties of Physical processes affecting plant growth.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Properties of Physical processes affecting plant growth: Outcome Check

Outcome check for Properties of Physical processes affecting plant growth: Describe the physical properties affecting plant growth.

1. Create a two-column table or worksheet for a real class task.
2. Use one correct heading, formula, save step, or formatting step.
3. Explain how the digital file can be checked for errors.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Environmental phenomena and related physics concepts: Lesson Opener

Learning focus: Environmental phenomena and related physics concepts

Country link: a safe Rwandan observation, model, data table, or investigation for Environmental phenomena and related physics concepts.

Progression focus: apply ideas to unfamiliar situations and justify decisions with evidence; connect evidence, diagrams, safety, and conclusions.

By the end of this topic, you should be able to:

- relate physics concepts with environmental phenomena

Inquiry questions:

- How can environmental phenomena and related physics concepts help you solve a real problem in Physics?

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Environmental phenomena and related physics concepts: Learn

Environmental phenomena and related physics concepts is learned by turning the topic into a clear, safe, evidence-based science task.

Core idea:

- Name the science idea before starting: structure, process, material, force, energy, organism, environment, or system.
- Use a diagram, data table, labelled model, safe observation, or teacher-approved investigation.
- Record what is seen or measured, not what is guessed.
- Explain the result using the vocabulary from this topic.
- Finish with one conclusion and one question for further study.

Key words:

- Environmental phenomena and related physics concepts
- Environmental
- phenomena
- related
- physics
- concepts
- concept
- observation

Environmental phenomena and related physics concepts: Worked Example

Investigation model for Environmental phenomena and related physics concepts: Turn the topic into a safe, clear classroom investigation or model.

1. Question: write one question that can be answered by observing, comparing, classifying, measuring, modelling, or reading data.
2. Materials: list safe classroom materials, diagrams, samples, cards, tables, or teacher-approved apparatus.
3. Method: do one step at a time and keep one condition fair where comparison is needed.
4. Safety: name any heat, glass, sharp tool, chemical, hygiene, electrical, or body-safety rule before starting.
5. Record: use a labelled table, diagram, or sentence notes.
6. Conclusion: state what the evidence shows and one limit of the investigation.

Answer check: a strong response names the concept, uses correct vocabulary, records evidence, and does not claim more than the evidence supports.

Environmental phenomena and related physics concepts: Vocabulary And Study Skill

Use these words and study moves before you practise Environmental phenomena and related physics concepts. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Environmental phenomena and related physics concepts
- Environmental
- phenomena
- related
- physics
- concepts
- concept
- observation

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Environmental phenomena and related physics concepts without a clear example, method, or evidence.

Environmental phenomena and related physics concepts: Guided Activity

Work with a partner or small group.

1. Read the inquiry question again.
2. Choose one real situation linked to a safe Rwandan observation, model, data table, or investigation for Environmental phenomena and related physics concepts.
3. Decide what evidence, method, vocabulary, tool, text, map, diagram, movement, or example is needed.
4. Complete the task together and explain each step aloud.
5. Compare your answer with another group.
6. Improve one part after feedback.

Group role check: one learner reads the task, one records ideas, one checks vocabulary or working, and one reports the answer.

Environmental phenomena and related physics concepts: Practice And Correction

Practice for Environmental phenomena and related physics concepts:

Main outcome: relate physics concepts with environmental phenomena.

1. Write one safe investigation question.
2. Name materials and observations needed.
3. Use the evidence to write a conclusion.

Answer check:

A strong science answer names the idea, uses correct units or labels where needed, records evidence, and writes a conclusion that follows from the evidence.

Safety check: use only teacher-approved materials and never test mains electricity, heat, sharp tools, or chemicals without direct supervision.

Environmental phenomena and related physics concepts: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Environmental phenomena and related physics concepts.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Environmental phenomena and related physics concepts: Outcome Check

Outcome check for Environmental phenomena and related physics concepts: Relate physics concepts with environmental phenomena.

1. State a safe investigation question.
2. Name the evidence needed.
3. Write a conclusion from the evidence.

Success criteria:

- evidence is specific
- safety is considered
- conclusion uses the evidence

Graphs of linear motion: Review Clinic 1

Use this review clinic to strengthen Physics without copying earlier answers.

1. Choose one idea from Graphs of linear motion that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Friction force and Newton's laws of motion: Review Clinic 2

Use this review clinic to strengthen Physics without copying earlier answers.

1. Choose one idea from Friction force and Newton's laws of motion that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Applications of atmospheric pressure: Review Clinic 3

Use this review clinic to strengthen Physics without copying earlier answers.

1. Choose one idea from Applications of atmospheric pressure that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Renewable and non-renewable energy sources: Review Clinic 4

Use this review clinic to strengthen Physics without copying earlier answers.

1. Choose one idea from Renewable and non-renewable energy sources that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Glossary

Use this glossary to revise important words in Physics.

- Graphs of linear motion: Explain this term using an example from the book, your school, or your community.
- Friction force and Newton's laws of motion: Explain this term using an example from the book, your school, or your community.
- Applications of atmospheric pressure: Explain this term using an example from the book, your school, or your community.
- Renewable and non-renewable energy sources: Explain this term using an example from the book, your school, or your community.
- Heat transfer and quantity: Explain this term using an example from the book, your school, or your community.
- Laws of thermodynamics: Explain this term using an example from the book, your school, or your community.
- Electrical power transmission: Explain this term using an example from the book, your school, or your community.
- Electric field intensity: Explain this term using an example from the book, your school, or your community.
- House electric installation: Explain this term using an example from the book, your school, or your community.
- Basic alternating current circuits: Explain this term using an example from the book, your school, or your community.
- Refraction of light: Explain this term using an example from the book, your school, or your community.
- Telecommunication Channels: Explain this term using an example from the book, your school, or your community.
- Properties of Physical processes affecting plant growth: Explain this term using an example from the book, your school, or your community.
- Environmental phenomena and related physics concepts: Explain this term using an example from the book, your school, or your community.

Add five more words from your lessons and write your own meanings.

Answer And Teacher Notes

A strong Physics answer states the idea, uses correct scientific vocabulary, records observations or data, includes units or labels where needed, and writes a conclusion that follows from evidence. Safety rules always come first. If evidence is missing, repeat the observation safely or explain what evidence would be needed.

Final Project

Create a final project for Physics that shows what you can now do independently.

Your project must include:

1. A clear title.
2. A real-life problem, question, text, performance, investigation, design, or worked task.
3. Evidence from at least three lessons in this book.
4. A drawing, table, map, chart, model, dialogue, paragraph, performance plan, practical product, or worked solution.
5. A correction note showing one improvement after feedback.
6. A short reflection explaining what you learned, what was difficult, and what you will practise next.

Presentation check: keep the work neat, label important parts, use correct vocabulary, and be ready to explain your choices to a classmate or teacher.